This professional **SX1276 LoRa Wireless Transceiver Module** is a high-performance, long-range communication solution designed for the Internet of Things (IoT). Utilizing Semtech's patented LoRa modulation technology, this module achieves extreme sensitivity and robust anti-interference performance, making it the ideal choice for smart meters, agricultural monitoring, and remote industrial control where traditional wireless protocols fail.

---

1. Product Info

The SX1276 module operates in the sub-GHz bands (868MHz/915MHz) and excels in low-power, long-range data transmission. By employing spread-spectrum communication, it can achieve a range of several kilometers in line-of-sight conditions while maintaining ultra-low current consumption.

> **LoRa Modulation:** Provides significantly higher link budget and range compared to FSK or GFSK.
>
> **High Sensitivity:** Capable of receiving signals as low as -148 dBm, ensuring reliable connections in challenging environments.

Anti-Interference: Excellent selectivity and blocking performance against nearby interfering signals.

Standard Interface: Uses a 4-wire SPI interface for easy integration with Arduino, ESP32, STM32, and Raspberry Pi.

Compact Footprint: Stamp-hole (castellated) design allows for both breadboard prototyping and high-volume SMT production.

---

2. Safety & Warnings

Antenna Requirement: NEVER power up or transmit without the antenna securely soldered. Operating the RF chip into an open circuit (no load) can cause reflected power to permanently damage the internal power amplifier.

Voltage Limit: This is a **3.3V** logic device. The maximum input voltage is **3.7V**. Connecting to 5V power or high-voltage logic without level shifters will destroy the module.

RF Exposure: During high-power transmission, maintain a safe distance from the antenna to avoid unnecessary RF exposure.

ESD Protection: RF components are extremely sensitive to Electrostatic Discharge (ESD). Handle the module in an anti-static environment.

---

3. Instructions for Use

Pinout Definition:

VCC: Connect to a stable 3.3V DC source.

GND: System Ground.

SCK / MISO / MOSI / NSS: Standard SPI communication pins.

DIO0 - DIO5: Digital I/O pins for interrupts (e.g., Packet Received, Tx Done). DIO0 is typically used for most LoRa libraries.

RESET: Reset pin (Active Low).

ANT: Antenna connection point.

Antenna Installation:

Solder the included spring antenna to the **ANT** pad.

Ensure the antenna is positioned vertically and kept away from large metal objects or high-current power traces for optimal range.

Software Setup:

For Arduino users, the **"RadioHead"** or **"LoRa" (by Sandeep Mistry)** libraries are highly recommended. Ensure the frequency is set in the code to match your hardware (868MHz or 915MHz) and local regulations.

---

4. Technical Specifications

Feature	Specification
RF Chip	Semtech SX1276
Frequency Range	868MHz / 915MHz (Software Selectable)
Modulation	LoRa / FSK / GFSK / OOK
Max Transmit Power	+20 dBm (100mW)
Receiving Sensitivity	Up to -148 dBm
Operating Voltage	1.8V – 3.7V DC (Typical 3.3V)
Emission Current	~120mA (+20dBm)
Receiving Current	~12mA
Sleep Current	< 1μA
Interface	SPI

---

5. Maintenance & Storage

Environment: Store in a dry, anti-static bag at room temperature. Moisture can lead to oxidation of the gold-plated pads.

Soldering: Use a temperature-controlled iron at approximately $320^\circ C$. Avoid excessive heat on the RF shield or the crystal oscillator (the small silver component).

Cleaning: If necessary, clean the pads with a small amount of isopropyl alcohol and a lint-free swab. Ensure the module is completely dry before applying power.

---

6. Disposal & Recycling

As an electronic RF component, this module contains specialized semiconductor materials and should not be discarded with standard household waste. Please follow your local **E-Waste** regulations and take the module to a certified recycling facility. This ensures that the silicon, copper, and fiberglass materials are processed safely and responsibly.